

Muhammad Gohar JAVED

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🎓 EDUCATION

December 2024 September 2022	Master of Science in Electrical and Computer Engineering University of Alberta, Canada ➤ CGPA : 4.00 / 4.00
July 2020 September 2016	Bachelor of Electrical Engineering National University of Sciences and Technology (NUST), Pakistan ➤ CGPA : 3.84 / 4.00 Rank : top 5%

📁 WORK EXPERIENCE

Aug 2024 May 2024	Machine Learning Advisor Amii - Alberta Machine Intelligence Institute, Edmonton AB, Canada ➤ Helped startups scale their Machine Learning capabilities through c-suite coaching sessions, as part of the "Technical Advisement - Client Coaching" work integrated learning opportunity.
Aug 2023 May 2023	Associate Machine Learning Developer AltaML, Edmonton AB, Canada ➤ Developed Proof-of-Concept for a Fleet Maintenance Prediction product for the City of Calgary, using historical maintenance records, vehicle telemetry data and road and weather condition indicators.
Present September 2022	Graduate Research Fellow Faculty of Engineering, University of Alberta, Canada ➤ Working under the supervision of Dr. Li Cheng at Vision and Learning Lab on the development of cutting edge Generative AI techniques for 3D Human Motion and Interaction Synthesis.
April 2023 May 2022	Machine Learning Researcher Hotpot AI, Los Altos CA, USA (Remote) ➤ Led the upgradation of the Object Removal and Background Removal projects. ➤ Improved mean absolute error in Background Removal by 5 times on a custom-curated test dataset, leading to several B2B contracts worth tens of thousands of dollars. ➤ Applied a novel Generative AI based controllable inpainting technique for Object removal.
May 2022 October 2020	Data Scientist Teradata Global Consulting, Islamabad, Pakistan ➤ NOV United States - Designed a computer vision solution to automate the segmentation of drill-bits from 3D point clouds for damage inspection. Improved the accuracy of the existing solution from 40% to 87% , leading the project from PoC (Proof of Concept) stage to MVP stage (Minimum Viable Product). ➤ Resona Bank Japan - Developed Retail Vision as-a-Service (RVaaS), an intelligent computer vision service to extract hidden insights from video streams and analyze in-store customer behavior at retail locations. Trained and deployed object detection, multi-object tracking and perspective transformations to identify patterns of human activity.

☰ SKILLS

Programming and Scripting :	Python, C, C++, Bash, Batch and MATLAB.
Machine Learning :	PyTorch, Tensorflow, ONNX, Scikit-learn, OpenCV and SQL.
2D & 3D Computer Vision :	Python, OpenCV, Open3D, Trimesh and Matlab.
Data Analysis and Visualization :	Python, Pandas, Seaborn, Matplotlib, Plotly and SQL.
Software Development Tools :	Git, Github, Docker and Jupyter.
Cloud Technologies :	AWS Elastic Container Service (ECS), ECR, EC2, S3, Azure Cognitive Services and VMs.
Embedded Systems	ARM, Arduino, Xilinx FPGAs and Huawei Atlas.

PUBLICATIONS & PROJECTS

INTERMASK : 3D HUMAN INTERACTION GENERATION VIA COLLABORATIVE MASKED MODELLING

M. GOHAR JAVED, C. GUO, L. CHENG, X. LI

Arxiv Preprint 2024 - [PROJECT WEBPAGE](#)

A novel framework for human interaction generation using collaborative masked modeling in the discrete space, which explicitly models spatio-temporal dependencies within and between the interacting individuals.

MoMASK : GENERATIVE MASKED MODELING OF 3D HUMAN MOTIONS

C. GUO*, Y. MU*, M. GOHAR JAVED*, S. WANG, L. CHENG *Co-first Authors

CVPR 2024 - [PROJECT WEBPAGE](#)

A novel masked modeling framework for text-driven 3D human motion generation, using hierarchical motion tokenization and masked bidirectional transformers.

QUANTYOLO : A HIGH-THROUGHPUT & POWER-EFFICIENT OBJECT DETECTION NETWORK FOR RESOURCE CONSTRAINED UAVS

M. GOHAR JAVED, M. RAZA, M. M. GHAFAR, C. WEIS, F. SHAFAT, N. WEHN

DICTA 2021 - [PAPER URL](#)

A fully-binarized single-shot object detection network with 1 bit weights, 4 bit activations, and only 9 conv layers for resource and power constraint devices in edge applications.

REVISITING FUZZY TILING ACTIVATION AND HOW TO SET ITS HYPERPARAMETERS

Reinforcement Learning Course Project, UAlberta - [PAPER URL](#) | [CODE REPO](#)

Addresses the sensitivity of Fuzzy Tiling Activation (FTA) to its tiling bounds by using scaled Tanh to limit the range of its input.

MEASURING RACIAL BIAS IN IMAGE CAPTIONING MODELS USING COUNTERFACTUALS

Privacy and Fairness in ML Course Project, UAlberta - [PAPER URL](#) | [CODE REPO](#)

A method to measure racial bias in image captioning models using counterfactual images, that are changed minimally in only the sensitive attribute.

INVITED TALKS

2024 **Cohere Community Talks** - MoMask : Generative Masked Modeling of 3D Human Motions. ([Recording URL](#))

HONORS AND AWARDS

- 2023 **CVPR 2023 Student Grant** - Received a grant to attend the 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition covering conference registration, travel, and accommodation expenses.
- 2022 **UAlberta Graduate Research Fellowship** - Secured a fully funded Masters in Research position at the University of Alberta, Faculty of Engineering. CAD 19,500 / year award.
- 2021 **Teradata Spot Award** - For exceeding expectations in the completion and delivery of Vision Analytics Proof-of-Concept for Resona Bank Japan.
- 2019 **University of Tokyo Summer Internship Program (UTSIP 2019)** - Selected with a 6% acceptance rate with an award of JPY 110,000 and accommodation benefits.
- 2018-2020 **Academic Excellence Award** - Achieved a perfect GPA of 4.00/4.00 in 4 semesters out of 8.
- 2016 **International Physics Olympiad Switzerland** - Selected among the top 5 students to represent Pakistan after 3 extensive examination camps.

CERTIFICATIONS

- 2022 **Reinforcement Learning Specialization** - University of Alberta (Coursera)
- 2021 **AI Engineer Associate** - Microsoft Azure
- 2020 **Deep Learning Specialization** - deeplearning.ai (Coursera)
- 2020 **Advanced SQL** - Teradata
- 2020 **Machine Learning Fundamentals** - AWS